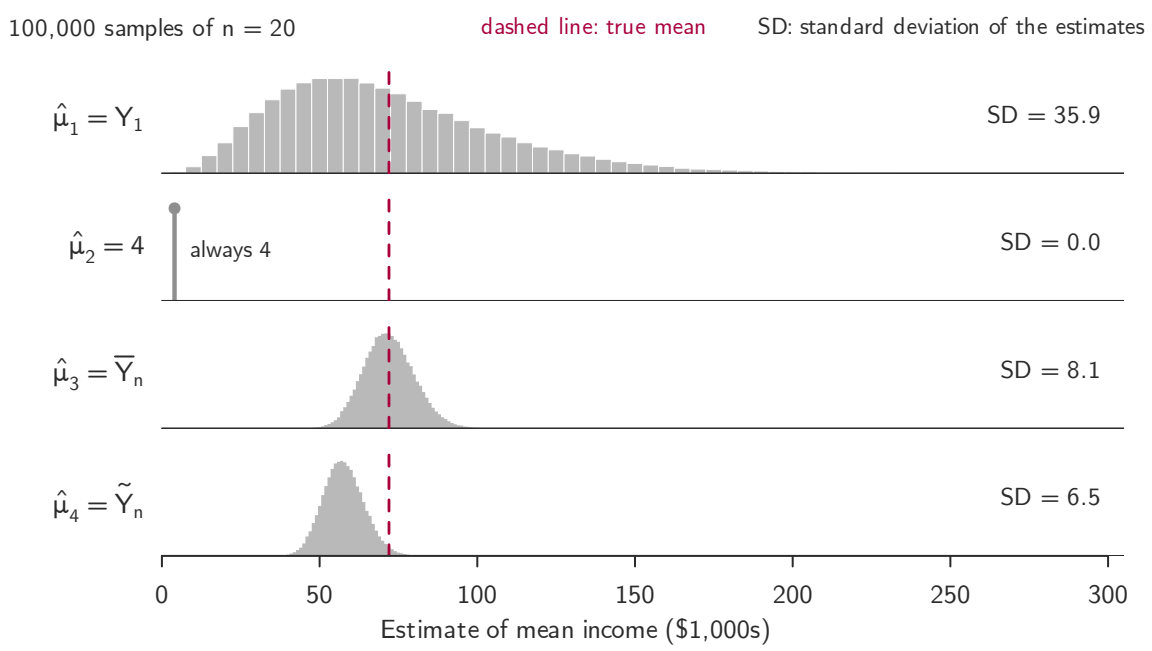
100,000 samples of n = 20
dashed line: true mean
SD: standard deviation of the estimates
$\hat{\mu}_1 = Y_1$
SD = 35.9
$\hat{\mu}_2 = 4$
always 4
SD = 0.0
$\hat{\mu}_3 = \overline{Y}_n$
SD = 8.1
$\hat{\mu}_4 = \tilde{Y}_n$
SD = 6.5
0
50
100
150
200
250
300
Estimate of mean income ($1,000s)